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MILITARY SPECIFICATION

ORAL INFLATION ASSEMBLIES, SURVIVAL EQUIPMENT, INFLATABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for two types of inflatable survival equipment oral inflation assemblies, three classes of oral inflation valves and five styles of oral inflation tubes.

1.2 Classification. The oral inflation assemblies shall be of the following types, the oral inflation valves shall be of the following classes and the oral inflation tubes shall be of the following styles, as specified (see 6.2.1b).

1.2.1 Types. The oral inflation assemblies shall be of the following types:

Type I - Cemented bond between the oral inflation tube and valve
Type II - Clamping of the oral inflation tube to the oral inflation valve

1.2.2 Classes. The oral inflation valves shall be of the following classes:

Class 1 - Standard size (used with styles A, B, C, and E oral inflation tubes).
Class 2 - Stubby standard size (used with styles A, B, C, and E oral inflation tubes).
Class 3 - Miniature size (used with style D and E oral inflation tube).

1.2.3 Styles. The oral inflation tubes shall be of the following styles:

Style A (standard)	- 62A119D1
Style B (standard corrugated)	- 62A119D1
Style C (low profile)	- 52B6586-1
Style D (low profile miniature)	- 52B6586-2
Style E (heat sealable-standard)	- 62A119D1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

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MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-A-5540	-	Adhesive, Polychloroprene.
MIL-R-6855	-	Rubber, Synthetic, Sheets, Strips, Molded, or Extruded Shapes.
MIL-T-46072	-	Tube, Round, Copper Alloy Numbers 330, 331, 332, and 370 (Leaded Brass), Seamless.

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking for U.S. Military Property
MIL-STD-889	-	Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

62A119D1	-	Oral Inflation Assemblies, Inflatable Survival Equipment.

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AIR FORCE

52B6586 Tube - Oral Inflation, Pneumatic Survival Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1c and 6.2.1d), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials and components. The materials and components shall conform to the applicable specifications, standards and drawings as listed or required herein. Unless otherwise specified, the manufacturer shall certify that the materials and components, except for the metallic parts used in the construction of the oral inflation valve, have been manufactured not more than 18 months prior to the date of delivery of the oral inflations assemblies, oral inflations valves or oral inflation tubes, as applicable (see 6.2.2).

3.3 Design and construction. The inflatable survival equipment oral inflation assembly shall consist of an oral inflation valve inserted within the non-flange end of the oral inflation tube. The tube shall be bonded or clamped to the valve, as applicable.

3.3.1 Oral inflation valve. The oral inflation valve shall be a spring-loaded self-closing type with a rubber-covered mouthpiece, for the classes 1 and 3 valves, at one end and a body with a serrated outlet, for the oral inflation tube, at the other end. The mouthpiece of the class 2 valve shall not be covered. The valve shall be capable of being locked in the closed position by means of a knurled ring (mouthpiece cannot be depressed). The serrated outlet on the classes 1 and 2 valves shall be designed for use with the 3/8 inch internal diameter of the oral inflation tubes, styles A, B and C. The class 3 valve outlet shall be designed for use with the 3/16 inch internal diameter of the oral inflation tube, style D. The diametral dimensions of the serrated portion shall be such that there shall be an interference fit between the inside diameter of the applicable oral inflation tube and the serrated outlet of the oral inflation valve. The body portion of the valve shall be fabricated from metal and the mouthpiece portion of the classes 1 and 3 valves may be fabricated from metal or an acetal resin "Delrin 500," as manufactured by E. I. DuPont DeNemours and Company, Incorporated, Wilmington, DE 19898, or equivalent. The color of the acetal resin shall be black. The mouthpiece of the class 2 valve shall be fabricated from the aforementioned acetal resin. The valve shall be so constructed that no part shall become detached during usage. The configuration of the oral inflation valves shall conform to figures 1 and 2, as applicable.

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3.3.1.1 Metals. The metals used in the oral inflation valve shall be of the corrosion resistant type or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the inflation valve may be subjected, when in storage or during normal service life.

3.3.1.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

3.3.1.1.2 Protective treatment. When materials are used in the construction of the inflation valve that are subject to deterioration, when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings which might crack, chip, flake, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.3.1.2 Rubber components. The rubber components used in the construction of the oral inflation valves shall conform to MIL-R-6855, class 2, grade 50, 60, 70 or 80. There shall be an interference fit between the outside diameter of the metal or plastic mouthpiece, as applicable, classes 1 and 3 oral inflation valves, and the inside diameter of the rubber mouthpiece. The bottom end of the rubber mouthpiece shall be flush against the ridge on the mouthpiece (see figure 1, note 2). The class 2 valve does not have a rubber mouthpiece.

3.3.2 Oral inflation tube.

3.3.2.1 Rubber oral inflation tube. The rubber oral inflation tube shall be a single molded construction with or without corrugations, as applicable. The oral inflation tube shall be fabricated from synthetic rubber conforming to MIL-R-6855, class 2, grade 60 for styles A, C and D, and grade 50 for style B. The oral inflation tube shall conform to Drawing 62A119D1 for styles A and B and Drawing 52B6586-1 and -2, for styles C and D, as applicable. The base, if desired, may be lightly buffed, on both sides, prior to being cemented to the life preserver, raft or any other kind of survival equipment.

3.3.2.2 Heat sealable oral inflation tube. The heat sealable oral inflation tube, style E, shall be Halkey-Roberts, Division of Walter Kidde and Company, Paramus, NJ 07652, Part No. 727-ACU-43, or equivalent (see Drawing 62A119D1-101).

3.3.3 Oral inflation assembly. The oral inflation assembly shall be formed by bonding or clamping the oral inflation tube to the valve.

3.3.3.1 Bonding of the oral inflation tube to the oral inflation valve. The serrated portion of the oral inflation valve shall be inserted and cemented within the non-flange end of the styles A, B or C oral inflation tubes for 3/4 inch and for 7/16 inch within the style D oral inflation tube. The surface to which the adhesive is to be applied shall be thoroughly cleaned with a suitable solvent in such a manner that the dusting materials (zinc stearate or talc) or any other surface contaminant is removed. The surface shall be clean prior to cementing. Care should be exercised to insure that the oral inflation tube and the elastomeric components of the oral inflation valve are not damaged in any way by prolonged

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exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive, and shall leave no residue. The adhesion shall be controlled to insure that old adhesive or adhesive which has partly or completely polymerized is not used. A fresh batch of the adhesive shall be used at least every eight hours during the course of manufactuer. Containers for the adhesive shall be free from congealed adhesive before being refilled. The adhesive shall not be allowed to remain in clots and upon drying or curing shall not cause the tube to shrink or pucker. The external portion of the oral inflation valve shall be free from adhesive. The adhesive shall conform to MIL-A-5540, class 1 or 2.

3.3.3.2 Clamping of the oral inflation tube to the oral inflation valve. The serrated portion of the oral inflation valve shall be inserted within the non-flange end of the styles A, B, C or E oral inflation tubes for 3/4 inch and for 7/16 inch within the style D oral inflation tube. The metal ring shall be inserted on the tube as in figure 3 and shall be crimped in place without damaging the oral inflation tube or the valve. The crimping shall securely attach the tube to the valve. The clamp shall be a seamless brass tubing conforming to MIL-T-46072, copper alloy number 330 or 332, temper drawn, and figure 3, as applicable.

3.4 Performance characteristics.

3.4.1 Oral inflation valve. The following inspections are applicable to all the classes of the oral inflation valves.

3.4.1.1 Force to open. When inspected as specified in 4.6.2.1, the force required to depress the mouthpiece to its maximum amount of travel shall be 2-1/4 ± 1/4 pounds for the classes 1 and 3 valves and 1-1/4 ± 1/4 pounds for the class 2 valve. Upon release of the pressure, the mouthpiece shall return to its original position.

3.4.1.2 Flow. When inspected as specified in 4.6.2.2, the minimum flow of air through the valve shall conform to table I.

3.4.1.3 Leakage. When inspected as specified in 4.6.2.3, the valve shall conform to table II. Neither the gasket nor any component of the oral inflation valve shall be dislodged during the leakage at 21°C (70°F) inspection. The leakage at 71°C (160°F) shall be performed in sequence with the leakage at -54°C (-65°F). The leakage inspection at 71°C (160°F) shall be performed first and when completed, the leakage inspection at -54°C (-65°F) shall then be determined.

3.4.1.4 Weight. When determined as specified in 4.6.2.4, the maximum weight of each valve shall be as specified in table III.

3.4.1.5 Strength of assembly. When inspected as specified in 4.6.2.5, the valve shall withstand a load of 75 pounds without the mouthpiece separating from the body, breaking or damage to any component or stretching the point of attachment of the mouthpiece to the body. The dial indicator shall return to within 0.010 inch of zero. The mouthpiece shall not separate from the body when the valve is locked in the closed position by the knurled ring. The valve shall then be inspected and conform to the leakage requirements at 21°C (70°F) for 1/2 pound per square inch gage (psig) pressure (see 3.4.1.3).

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3.4.1.6 Endurance. When inspected as specified in 4.6.2.6, the mouthpiece shall not separate from the body nor shall there be any evidence of damage to any component. The valve shall then be inspected and conform to the requirements for force to open, 3.4.1.1, and 1/2 psig leakage at 21°C (70°F), 3.4.1.3.

3.4.1.7 Salt spray. When inspected as specified in 4.6.2.7, the oral inflation valve shall show no evidence of corrosion or damage. The mouthpiece and the knurled ring shall not become frozen. The mouthpiece shall be capable of being depressed to its maximum amount of travel without any difficulty and the knurled ring shall operate freely. The valve shall then be inspected and conform to the requirements for force to open, 3.4.1.1. flow, 3.4.1.2, and 1/2 psig leakage at 21°C (70°F), 3.4.1.3.

3.4.2 Oral inflation tube. The following inspections are applicable to all the styles of the oral inflation tubes:

3.4.2.1 Porosity. The oral inflation tube, when inspected for porosity as specified in 4.6.3.1, shall not leak or burst.

3.4.2.2 Breaking strength. The oral inflation tube shall have a minimum breaking strength of 50 pounds, when inspected as specified in 4.6.3.2.

3.4.3 Oral inflation assembly. The following inspections are applicable to all the types of the oral inflation assemblies:

3.4.3.1 Porosity. The oral inflation assembly, when inspected for porosity as specified in 4.6.4.1, shall not leak and the oral inflation tube shall not burst. The oral inflation valve shall operate without difficulty, shall shut off the flow of air completely, and when applicable, shall not be blocked or affected by excess adhesive.

3.4.3.2 Breaking strength of the oral inflation tube and strength of attachment of the oral inflation tube to the valve. When inspected as specified in 4.6.4.2, the breaking strength of the oral inflation tube shall be not less than 50 pounds and the strength of attachment of the oral inflation tube to the valve shall be not less than 50 pounds.

3.5 Markings. Markings shall be in accordance with MIL-STD-130.

3.5.1 Oral inflation valve. The markings on the knurled ring and stem ridge shall be indent stamped. The characters shall be 1/16 inch in height.

3.5.1.1 Knurled ring. The knurled ring shall contain the following markings:

Manufacturer's name or trademark
Manufacturer's part number
Last two digits of the year of manufacture

3.5.1.2 Body ridge. If desired, the body ridge may be marked with the name of the manufacturer or trademark instead of the knurled ring.

3.5.2 Oral inflation tube. Unless otherwise specified, the tube portion shall be marked with permanent yellow or white letters and numerals, which shall be thoroughly

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dry prior to packaging. If desired, the markings may be molded characters of the same color as the tube or a combination of marks and molded characters. The markings shall be as follows:

Manufacturer's name or trademark
Date of cure (month and year)

3.6 Dimensions. The dimensions of the oral inflation valve, oral inflation tube, clamp, length of insertion of the oral inflation valve into the oral inflation tube and the distance of the clamp from the end of the oral inflation tube shall conform to 3.3.1.2, 3.3.3.1, 3.3.3.2, figures 1, 2, and 3, and Drawings 62A119D1 and 52B6586, as applicable, when determined as specified in 4.6.5.

3.7 Workmanship. The oral inflation valves, tubes or assemblies, as applicable, shall not be deformed or misaligned nor contain any rough surface, nick, burr, sharp edge, crack, spot, stain, corrosion or scale, pit, dent, embedded foreign matter, loose or detached component. No component of the oral inflation assemblies shall contain any non-specified hole, tear, cut, patch, mend, burn or abraded area. Because of the emergency and life support use of this equipment, the importance of providing a product of uniform excellent quality cannot be overemphasized. The oral inflation valves, tubes or assemblies, as applicable, shall be uniform in quality and free from irregularities or defects which could adversely affect performance, reliability or durability. The oral inflation valves, tubes or assemblies, as applicable, shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit six oral inflation valves of each class, six oral inflation tubes of each style, or six oral inflation assemblies of each type, class of valve and style of tube specified in the contract or order, as applicable. The samples shall be representative of

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the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these oral inflation valves, tubes or assemblies, as applicable, from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1c). Approval of the first article inspection samples or the waiving of the first article inspection does not exempt the requirements for performing the quality conformance inspection. The first article samples shall be forwarded as specified in the contract or order (see 6.2.1d).

4.3.2 First article sample approval. Upon completion of the first article inspection program, pertinent comments and recommendations will be forwarded by the Government activity responsible for conducting the inspection program (see 6.2.1d) to the contracting officer. One approved oral inflation valve, tube or assembly, as applicable, will be returned to the manufacturer for use in monitoring production. The other five oral inflation valves, tubes or assemblies, as applicable, will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.3.3 First article inspection. The first article inspection of the oral inflation valves, tubes or assemblies, as applicable, shall consist of examinations and tests specified in table IV.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the oral inflation valves, tubes or assemblies, as applicable.
Dimensional check of the oral inflation valves, tubes or assemblies, as applicable.
Force to open.
Flow.
Leakage at 21°C (70°F).
Weight.
Strength of assembly.
Endurance.
Porosity of the oral inflation tube.
Breaking strength of the oral inflation tube.
Porosity of the oral inflation assembly.
Breaking strength of the oral inflation tube and strength of attachment of the oral inflation tube to the valve (oral inflation assembly).
Packaging.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Oral inflation valves. An inspection lot size shall be expressed in units of one oral inflation valve of one class made essentially under the same conditions and from the same materials. The sample unit shall be one oral inflation

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valve of one class.

4.4.1.1.2 Oral inflation tubes. An inspection lot size shall be expressed in units of one oral inflation tube of one style made essentially under the same conditions and from the same material. The sample unit shall be one oral inflation tube of one style.

4.4.1.1.3 Oral inflation assemblies. An inspection lot size shall be expressed in units of one oral inflation assembly of one type, class of valve and style of tube made essentially under the same conditions and from the same materials and components. The sample unit shall be one oral inflation assembly of one type, class of valve and style of tube.

4.4.1.1.4 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing oral inflation valves of one class, oral inflation tubes of one style or oral inflation assemblies of one type, class of valve and style of tube, as applicable, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing oral inflation valves of one class, oral inflation tubes of one style or oral inflation assemblies of one type, class of valve and style of tube, as applicable, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for examinations and tests of the oral inflation valves, tubes, or assemblies. The sample size, acceptance criteria, examinations and tests required for the oral inflation valves, tubes or assemblies shall be as specified in table V.

4.5 Inspection conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all inspections required by this specification shall be conducted at an atmospheric pressure of 760 mm (28 to 32 inches) of mercury and at a temperature of 25° ± 10°C (77° ± 18°F). When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument reading.

4.5.2 Inspection area and equipment. The area in which the oral inflation valves, tubes or assemblies, as applicable, are inspected shall be adequately protected to preclude damage to the units. The area and inspection equipment shall be free of sharp or rough edges, burrs, protrusions, etc., which will cut, tear or damage the oral inflation valves, tubes or assemblies, as applicable.

4.5.3 Air. When use of air is specified in an inspection, the air used shall not contain any oil or condensed water vapor.

4.5.4 Drying. Unless otherwise specified, the oral inflation valves, tubes or assemblies, as applicable, upon completion of the inspections, which require the oral inflation valves, tubes or assemblies, as applicable, to be immersed in a liquid, shall be dried in a manner that shall not affect the serviceability of the valves, tubes or assemblies, as applicable. The valves, tubes or assemblies, as applicable, may be dried at temperatures higher than ambient but shall not exceed 52°C (125°F). The oral inflation valves, tubes or assemblies, as applicable,

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shall only be dried for approximately two hours at temperatures of 38° to 52°C (100° to 125°F).

4.6 Inspection methods.

4.6.1 Visual examination.

4.6.1.1 Oral inflation valves, tubes or assemblies. Every oral inflation valve, tube or assembly, as applicable, shall be visually examined for critical defects to determine conformance to this specification. Every oral inflation valve, tube or assembly, as applicable, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and visually examined for minor defects to determine conformance to this specification. The oral inflation valve and tube shall be examined visually and checked dimensionally, when the valve and tube are a component of the oral inflation assembly and when they are acquired individually. The classification and list of defects, tables VI and VII, as applicable, shall be used to classify and enumerate the defects found.

4.6.1.2 Packaging. Each of the fully prepared shipping containers, containing oral inflation valves of one class, oral inflation tubes of one style or oral inflation assemblies of one type, class of valve and style of tube, as applicable, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, table VIII, shall be used to enumerate the defects found.

4.6.2 Oral inflation valve.

4.6.2.1 Force to open. The oral inflation valve, with the knurled ring in the open position, shall be placed into a suitable reasonably friction free apparatus, such as in figure 4. The apparatus shall be mounted or held stationary and pressure (force) shall be applied to the valve slowly and steadily. The force shall be continued until the mouthpiece is depressed to its maximum amount of travel. Care shall be taken not to introduce any error due to bottoming of the mouthpiece or by the application of pressure by shock. The highest value obtained shall be reported as the result for the force to open inspection (see 3.4.1.1). The capacity of the spring scale shall be approximately 10 pounds and shall be capable of at least measuring to the nearest ounce. The force shall be released and the valve shall be examined for the position of the mouthpiece (see 3.4.1.1).

4.6.2.2 Flow. The knurled ring shall be in the open position and the mouthpiece shall be fully depressed. The valve shall then be attached to the flow inspection apparatus, shown in figure 5, or any other suitable flow apparatus and air pressure (see 4.5.3), not greater than 13 inches of water, shall be applied internally, at the depressed mouthpiece (inlet side), for a minimum of 30 seconds. After the flow of air has stabilized, the amount of air flowing through the valve per minute shall be determined (see 3.4.1.2).

4.6.2.3 Leakage.

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4.6.2.3.1 At 21°C (70°F). The class 1 or 2 oral inflation valve shall be suitably restrained, at the serrated end, in a manner that shall not interfere with the passage of air through the valve nor damage the valve. The knurled ring shall be in the open position and the mouthpiece shall be fully depressed. An instantaneous blast of 225 psig air pressure (see 4.5.3) shall be applied internally, at the inlet side of the valve (depressed mouthpiece), by means of a blow gun having a discharge nozzle not greater than 1/8 inch diameter. While the pressure is applied, the outlet side of the valve (serrated portion) shall be observed as to whether the gasket or any other component of the valve is dislodged. Care should be taken so that in the event any component is dislodged, the observer will not be injured. Upon completion of the internal high pressure air application and observation, the air supply shall be turned off, the mouthpiece shall be released to its original position, and the knurled ring shall remain in the open position. The valve shall be submerged under water, at a temperature of 21° ± 2.8°C (70° ± 5°F), and air pressure (see 4.5.3) of 1/2 psig shall be applied internally, at the outlet side of the valve, for a minimum of 10 seconds. While the pressure is applied, the valve shall be observed for leakage. Upon completion of the aforementioned, internally applied air pressure of 25 psig (see 4.5.3) shall be substituted for the 1/2 psig pressure, for a minimum period of 10 seconds, and the valve again observed for leakage. The application of 225 psig air pressure is not applicable to the class 3 valve. The class 3 oral inflation valve shall only be inspected for leakage by the application of 1/2 and 25 psig air pressure, respectively, as described herein. Upon completion of the testing and examinations, the air supply shall be turned off. The valve shall then be dried in accordance with 4.5.4.

4.6.2.3.2 At 71°C (160°F). The knurled ring shall be in the open position. The valve shall be submerged under water, in a vertical position with the mouthpiece uppermost, at a temperature of 71° ± 2.8°C (160° ± 5°F), for 24 ± 1/4 hours. During the exposure, air pressure of 25 psig (see 4.5.3) shall be applied internally, at the outlet side of the valve. Any leakage, during this 24 hour period, shall be collected in the calibrated apparatus shown in figure 6 or any other suitable calibrated collection device that has been completely filled with water. Suitable means shall be used to hold the collection device stationary. The oral inflation valve shall be placed, in relation to the collection device, so that leakage, from any point of the valve, shall be collected, by the displacement of water, in the collection device, to determine conformance to 3.4.1.3. The valve shall be suitably restrained in a manner that shall not damage the valve nor interfere with this inspection. The 25 psig air pressure may be supplied by a manifold or other suitable device. Upon completion of the aforementioned, the knurled ring shall be moved to the closed position and this inspection shall be conducted for an additional 24 ± 1/4 hours. Upon completion of the second 24 hour exposure, the air supply shall be turned off and the valve shall be dried in accordance with 4.5.4. After a minimum of 16 hours drying time, the oral inflation valve shall be subjected to the leakage inspection at -54°C (-65°F), 4.6.2.3.3.

4.6.2.3.3 At -54°C (-65°F). The leakage procedure specified in 4.6.2.3.2 shall be adhered to with the following modifications:

a. The immersion medium shall be 100 percent isopropyl alcohol.
b. This inspection shall be conducted in an ambient temperature of -54° ± 2.8°C (-65° ± 5°F).

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c. This inspection shall be conducted with internally applied air pressure of 1/2 psig (see 4.5.3).

4.6.2.4 Weight. The weight of each valve shall be determined on any scale or balance capable of weighing at least to the nearest 0.1 of a gram.

4.6.2.5 Strength of assembly. The oral inflation valve, with the knurled ring in the open position, shall be securely inserted within the jaws of the strength of assembly inspection apparatus, as shown in figure 7 or a similar device, without any damage to any portion of the valve. The dial gage shall be capable of being read in 0.001 inch. The dial indicator shall be set to the zero position. A 75 pound load, tending to separate the valve, shall be applied for a minimum of 30 seconds. At the end of 30 seconds, the pressure shall be released and the dial indicator read to determine conformance with 3.4.1.5. The valve shall then be removed from the jaws and visually examined for separation of the mouthpiece from the body, breaking or damage to any components or stretching of the point of attachment of the mouthpiece to the body (see 3.4.1.5). Upon completion of the visual examination, the valve shall be locked in the closed position, by the knurled ring, with hand tightening. The knurled ring shall then be turned so that the valve is in the open position completing one cycle. The aforementioned shall be repeated for an additional two cycles. The valve shall then be visually examined to determine whether the mouth piece has separated from the body (see 3.4.1.5). Upon completion of the examination, the valve shall be subjected to the leakage inspection at 21°C (70°F) for 1/2 psig pressure, 4.6.2.3.1.

4.6.2.6 Endurance. The oral inflation valve, with the knurled ring in the open position, shall be subjected to 500 operating cycles. One cycle shall consist of going from the normal position to the mouthpiece depressed to its maximum amount of travel to the normal position. While the valve is being cycled, air pressure of 5 psig (see 4.5.3) shall be applied internally at the inlet side of the valve. The number of cycles per minute shall be between 15 and 20. Upon completion of the 500 cycles, the valve shall be subjected to the following inspections, in the order listed:

Force to open	4.6.2.1
Leakage at 21°C (70°F) for 1/2 psig pressure	4.6.2.3.1

4.6.2.7 Salt spray. The knurled ring shall be in the open position and the oral inflation valve shall be exposed to 96 hours of salt spray in accordance with FED-STD-151, method 811. The outlet side of the valve shall be plugged to prevent the entrance of salt spray. The valve shall be exposed in a horizontal position in a manner that the salt spray circulates completely around the valve. At the expiration of 96 hours exposure, the valve shall be rinsed with tap, distilled or demineralized water, the outer surface wiped dry and the plug shall then be removed from the outlet side. The valve shall be exposed to atmospheric conditions, 4.5.1, for 48 ± 1/2 hours, in a manner that shall permit the air to circulate completely around the valve. The valve shall then be examined for corrosion, freezing of the mouthpiece and knurled ring and other damage to the valve that would affect operation (see 3.4.1.7). Upon completion of the examination, the valve shall be subjected to the following inspections, in the order listed:

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Force to open	4 6.2.1
Flow	4.6.2.2
Leakage at 21°C (70°F) for 1/2 psig pressure	4.6.2.3.1

4.6.3 Oral inflation tube. When the oral inflation tube is purchased individually, the following inspections shall be conducted:

4.6.3.1 Porosity. The oral inflation tube shall be submerged under water, at a temperature of 21° ± 2.8°C (70° ± 5°F), and shall be subjected to an internal air pressure of 15 psig (see 4.5.3) for a minimum of 10 seconds. The pressure shall be applied internally at the flanged end of the tube and the opposite end shall be plugged. While the pressure is applied, the tube shall not burst and shall be examined for porosity (see 3.4.2.1). Upon completion of the examination and testing, the oral inflation tube shall be dried in accordance with 4.5.4.

4.6.3.2 Breaking strength. The oral inflation tube base shall be folded along its minor axis so that the fold line shall be perpendicular to the main axis of the tube (see figure 8). The folded base shall then be clamped in the lower jaw of a suitable inspection apparatus. No portion of the curved tubular section, near the base, shall be clamped in the jaw. The free end of the tube portion shall be clamped in the upper jaw for a distance of 3/4 inch. The rate of jaw separation, under no load, shall be 12 ± 1/2 inches per minute. The jaws shall be separated until failure occurs (see 3.4.2.2). Upon completion of this inspection, the oral inflation tubes inspected shall be discarded.

4.6.4 Oral inflation assembly. With the oral inflation tube cemented or clamped to the oral inflation valve, as applicable, the oral inflation assembly shall be subjected to the following inspections:

4.6.4.1 Porosity. The knurled ring of the oral inflation valve shall be in the open position (mouthpiece can be depressed to its maximum amount of travel). The oral inflation assembly shall be submerged under water, at a temperature of 21° ± 2.8°C (70° ± 5°F), and shall be subjected to an internal air pressure of 15 psig (see 4.5.3) for a minimum of 10 seconds. The pressure shall be applied internally at the flange end of the oral inflation tube. While the pressure is applied, the oral inflation tube shall not burst and the oral inflation assembly shall be examined for porosity of the tube and for leakage at the point of insertion of the valve into the tube due to ineffectual cementing or clamping or by damage to the tube or valve by the clamping of the oral inflation tube to the valve, as applicable. Upon completion of the examination, the mouthpiece of the oral inflation valve shall be depressed to its maximum amount of travel to allow air to flow through the valve to determine whether there is any difficulty in depressing the mouthpiece or whether the flow of air is restricted by excess adhesive or by damage due to the clamping of the oral inflation tube to the valve, as applicable. Upon completion of the examination for the flow of air through the mouthpiece, the mouthpiece shall be released to its original position to determine whether the valve operates without difficulty and that the flow of air is shut off completely. Upon completion of the examination and testing, the oral inflation assembly shall be dried in accordance with 4.5.4.

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4.6.4.2 Breaking strength of the oral inflation tube and strength of attachment of the oral inflation valve to the tube. The oral inflation tube base shall be folded along its minor axis so that the fold line shall be perpendicular to the main axis of the tube (see figure 8). The folded base shall then be clamped in the lower jaw of a suitable inspection apparatus and the mouthpiece of the valve shall be secured to the upper jaw. No portion of the curved tubular section, near the base, shall be clamped in the jaw. The rate of jaw separation, under no load, shall be 12 ± 1/2 inches per minute. The jaws shall be separated until failure occurs (see 3.4.3.2). Upon completion of this inspection, the oral inflation assemblies inspected shall be discarded.

4.6.5 Procedure for measuring. The oral inflation valve, tube or assembly, as applicable, shall be laid flat, without tension, on a smooth flat surface. The dimensions of the valve, tube or assembly, as applicable, shall be determined by a micrometer, ruler or any suitable measuring device.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2.1e).

5.1.1 Level A.

5.1.1.1 Oral inflation valves. Each oral inflation valve shall be cleaned in accordance with MIL-P-116, process C-1 and preserved and packaged in accordance with MIL-P-116, method 1C-3. No contact preservative is required. A paper label shall be inserted (see 5.3.1) and immediately after cleaning, each valve shall be suitably cushioned with cellulosic material conforming to PPP-C-843, type II, class A or B. The cushioned valve shall then be heat sealed within a clear polyethylene snug fitting bag of 0.00125 ± 20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and to the edges of the polyethylene bag. The closing heat seal seam shall be as close as possible to the edge of the bag. Prior to or during the sealing operation, excess air shall be exhausted to the extent necessary to permit packing of the required quantity in the shipping container. The oral inflation assembly shall not be deformed in the packaged condition nor shall the sealed bag be damaged. The bagged assemblies of one type, class of valve, and style of tube shall be packaged within a snug fitting fiberboard box conforming to PPP-B-636, type CF, domestic class, variety SW, grade 275. The weight of the contents in each package shall not exceed 40 pounds. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches. All the seams and joints shall be sealed with 3 inch minimum wide tape conforming to PPP-T-45, type III, grade A, B or C.

5.1.1.2 Oral inflation tubes. A paper label shall be inserted (see 5.3.1) and each oral inflation tube shall be heat sealed within a clear polyethylene snug fitting bag 0.00125 ± 20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and the edges of the polyethylene bag. The closing heat seal seam shall be as close as possible to the edge of the bag. Prior to or during the sealing operation, excess air shall be exhausted to the

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extent necessary to permit packing of the required quantity in the container. The oral inflation tube shall not be deformed in the packaged condition nor shall the sealed bag be damaged. The bagged oral inflation tubes of one style shall be packaged within a snug fitting fiberboard box conforming to PPP-B-636, type CF, domestic class, variety SW, grade 275. The weight of the contents in each package shall not exceed 40 pounds. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches. All seams and joints shall be sealed with 3 inch minimum wide tape conforming to PPP-T-45, type III, grade A, B or C.

5.1.1.3 Oral inflation assemblies. Each oral inflation valve shall be cleaned in accordance with MIL-P-116, process C-1 and preserved and packaged in accordance with MIL-P-116, method 1C-3. No contact preservative is required. A paper label shall be inserted (see 5.3.1) and immediately after cleaning, each valve and clamp shall be suitably cushioned with cellulosic material conforming to PPP-C-843, type II, class A or B. The oral inflation assembly shall then be heat sealed within a clear polyethylene snug fitting bag of 0.00125 ± 20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and to the edges of the polyethylene bag. The closing heat seal seam shall be as close as possible to the edge of the bag. Prior to or during the sealing operation, excess air shall be exhausted to the extent necessary to permit packing of the required quantity in the shipping container. The oral inflation assembly shall not be deformed in the packaged condition nor shall the sealed bag be damaged. The bagged assemblies of one type, class of valve, and style of tube shall be packaged within a snug fitting fiberboard box conforming to PPP-B-636, type CF, domestic class, variety SW, grade 275. The weight of the contents in each package shall not exceed 40 pounds. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches. All the seams and joints shall be sealed with 3 inch minimum wide tape conforming to PPP-T-45, type III, grade A, B or C.

5.1.2 Commercial. The oral inflation valves of one class, the oral inflation tubes of one style and the oral inflation assemblies of one type, class of valve and style of tube, as applicable, shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2.1e). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight. Each shipping container shall contain oral inflation valves of one class, oral inflation tubes of one style, and oral inflation assemblies of one type, class of valve and style of tube, as applicable.

5.2.1 Level A. The oral inflation valves, tubes or assemblies, packaged as specified in 5.1.1 through 5.1.1.3, as applicable, shall be packed as specified in 5.2.2, except that the fiberboard container shall be weather-resistant class, variety SW, grade V3c or V3s. In addition, each container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

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5.2.2 Level B. The oral inflation valves, tubes or assemblies, packaged as specified in 5.1.1 through 5.1.1.3, as applicable, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, type CF or SF, domestic class, variety SW, grade 275. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636.

5.2.3 Commercial. The packaged oral inflation valves, tubes or assemblies, as applicable, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special marking required by the contract or order, the interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the class of the valve, style of the tube, type of the assembly, class of valve and style of tube, as applicable, and the date of manufacture of the valve (see 3.5.1 through 3.5.1.2) and the cure date of the tube (see 3.5.2), as applicable.

5.3.1 Polyethylene bag. Prior to heat sealing, a white paper label with legible and durable black letters and numerals shall be placed inside each polyethylene bag (see 5.1.1.1, 5.1.1.2, or 5.1.1.3, as applicable). The label shall be placed so that the information on the label shall be capable of being read through the bag. The label shall contain the same information as specified in 5.3. The label shall also contain the specification number.

6. NOTES

6.1 Intended use. The oral inflation assemblies covered by this specification are intended to be used to orally inflate inflatable survival equipment such as life preservers, one man life rafts and anti-exposure coverall hoods.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class of the valve and style of the tube of the oral inflation assembly (see 1.2.1); class of the oral inflation valve (see 1.2.2); style of the oral inflation tube (see 1.2.3), as applicable; Government part number; and quantity desired.

c. Whether first article inspection is waived (see 4.3.1).

d. Name and address of the first article inspection laboratory (see 4.3.1); including the Government activity responsible for conducting the inspection program (see 4.3.2).

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e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

f. Whether any special markings are required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement	Applicable DID no.
3.2	Certificate of compliance for the age of materials and components	DI-E-2121
4.3.2	First article inspection reports	DI-T-5329

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should be a first production item. The first article should consist of six units. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 99

Review activities:
Navy - SH, MC
Air Force - 11

User activities:
Army - AV, ME

Preparing activity:
Navy - AS

(Project No. 4220-0280)

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TABLE I. Air flow (see 3.4.1.2).

Class	Minimum liters per minute
1	50
2	50
3	12

TABLE II. Leakage limits (see 3.4.1.3).

Temperature °C (°F)	Maximum leakage milliliters
21 (70)	none
71 (160)	none
-54 (-65)	none

TABLE III. Weight of oral inflation valves (see 3.4.1.4).

Class	Maximum weight grams
1	30.0
2	30.0
3	7.5

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TABLE IV. First article examinations and tests (see 4.3.3).

Inspection	Paragraph	
	Requirement	Method
Visual examination	3.3 through 3.3.3.2, 3.5 through 3.5.2, 3.7, figures 1, 2, and 3 and applicable drawings	4.6.1.1 and Table VI
Dimensional check	3.6	4.6.1.1, 4.6.5 and Table VII
Force to open	3.4.1.1	4.6.2.1
Flow	3.4.1.2	4.6.2.2
Leakage	3.4.1.3	4.6.2.3.1
Weight	3.4.1.4	4.6.2.4
Strength of assembly	3.4.1.5	4.6.2.5
Endurance	3.4.1.6	4.6.2.6
Salt spray	3.4.1.7	4.6.2.7
Porosity	3.4.2.1 or 3.4.3.1, as applicable	4.6.3.1 or 4.6.4.1, as applicable
Breaking strength	3.4.2.2 or 3.4.3.2, as applicable	4.6.3.2 or 4.6.4.2, as applicable

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TABLE V. Sample size, acceptance criteria, examinations and tests of the oral inflation valves, tubes and assembiles.

Inspection	Paragraph		Sample Size	Acceptance Criteria
	Requirement	Method		
Visual examination 1/	3.3 through 3.3.3.2, 3.5 through 3.5.2, 3.7, figures 1, 2 and 3 and Drawings 62A119D1 and 52B6586, as applicable	4.6.1.1 and table VI	Every oral inflation valve, tube or assembly for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defect and an acceptable quality level of 15 defects per 100 units for minor defects.
Dimensional check of the oral inflation valves, tubes or assemblies, as applicable 1/	3.6	4.6.1.1, 4.6.5 and table VII	Inspection level S-3	An acceptable quality level of 4.0 defects per 100 units
Force to open 2/	3.4.1.1	4.6.2.1	Every oral inflation valve	Reject all defective units
Flow 2/	3.4.1.2	4.6.2.2	Every oral inflation valve	Reject all defective units
Leakage at 21°C (70°F) 2/	3.4.1.3	4.6.2.3.1	Every oral inflation valve	Reject all defective units
Weight 2/	3.4.1.4	4.6.2.4	Inspection level S-2	An acceptable quality level of 1.5 defects per 100 units
Strength of assembly 2/	3.4.1.5	4.6.2.5	Inspection level S-2	An acceptable quality level of 1.5 defects per 100 units
Endurance 2/	3.4.1.6	4.6.2.6	Inspection level S-1	An acceptable quality level of 1.5 defects per 100 units
Porosity of the oral inflation tube 3/	3.4.2.1	4.6.3.1	Every oral inflation tube	Reject all defective units

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TABLE V. Sample size, acceptance criteria, examinations, and tests of the oral inflation valves, tubes, and assemblies. - Continued

Inspection	Paragraph		Sample size	Acceptance criteria
	Requirement	Method		
Breaking strength of the oral inflation tube 3/ 4/	3.4.2.2	4.6.3.2	Inspection level S-2	An acceptable quality level of 1.5 defects per 100 units
Porosity of the oral inflation assembly 5/	3.4.3.1	4.6.4.1	Every oral inflation assembly	Reject all defective units
Breaking strength of the oral inflation tube and strength of attachment of the oral inflation tube to the valve 5/ 6/	3.4.3.2	4.6.4.2	Inspection level S-2	An acceptable quality level of 1.5 defects per 100 units
Packaging 7/	Section 5	4.6.1.2 and table VIII	Inspection level S-2	An acceptable quality level of 2.5 defects per 100 units

1/ This inspection is applicable to the oral inflation valves and tubes, when they are purchased individually and when they are components of the oral inflation assemblies.
2/ This inspection is applicable only to the oral inflation valve. This inspection is applicable when the oral inflation valves are purchased individually and when they are a component of the oral inflation assemblies.
3/ This inspection is applicable only to the oral inflation tubes when they are purchased individually.
4/ Upon completion of this inspection, the inspected oral inflation tubes shall be discarded and shall not be part of the quantity to be delivered under the contract or order.
5/ This inspection is applicable only to the oral inflation assemblies.
6/ Upon completion of this inspection, the inspected oral inflation assemblies shall be discarded and shall not be part of the quantity to be delivered under the contract or order.
7/ This inspection is applicable to the oral inflation assemblies and also to the oral inflation valves and tubes when they are purchased individually.

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TABLE VI. Classification of defects for the visual examination of the oral inflation valves, tubes, and assemblies (see 4.6.1.1).

Defects	
Critical	Minor
1. Any threads missing or stripped.	201. Knurling not uniform and continuous on the locking ring or the entire surface of the locking ring knurled.
2. Movement of the knurled locking ring interfered with 3/	
3. Not specified type, class, or style	202. Color of any component not as specified 2/
4. Any surface rough, misaligned or containing nicks, sharp edges cracks, burrs, scale, pit, dent, sliver or embedded foreign matter.	203. Any marking missing, incorrect, illegible, incomplete, not of a permanent nature, improperly located, or not the required color.
5. Any component missing, malformed, corroded, or otherwise damaged.	
6. Any component loose, detached, not attached as specified or otherwise not securely retained.	
7. Any non-specified hole, cut, tear, repair or burn.	
8. Oral inflation tube deformed, bent or not a single molded construction.	
9. Finish missing or improperly performed.	
10. Any component, component part, or required operation omitted, not as specified; or any operation improperly performed, not herein classified. 1/	

1/ The defect shall be classified critical when it seriously affects function or service ability; otherwise it shall be classified minor.

2/ The defect shall be classified minor when it does not seriously affect function or serviceability; otherwise it shall be classified critical.

3/ Movement of the knurled locking ring shall be verified by turning to lock the valve in the closed position and then by reversing completely so that the mouthpiece can be fully depressed to allow passage of air. This procedure shall be repeated a minimum of three times to insure that movement of the knurled ring and valve operation is in accordance with requirements.

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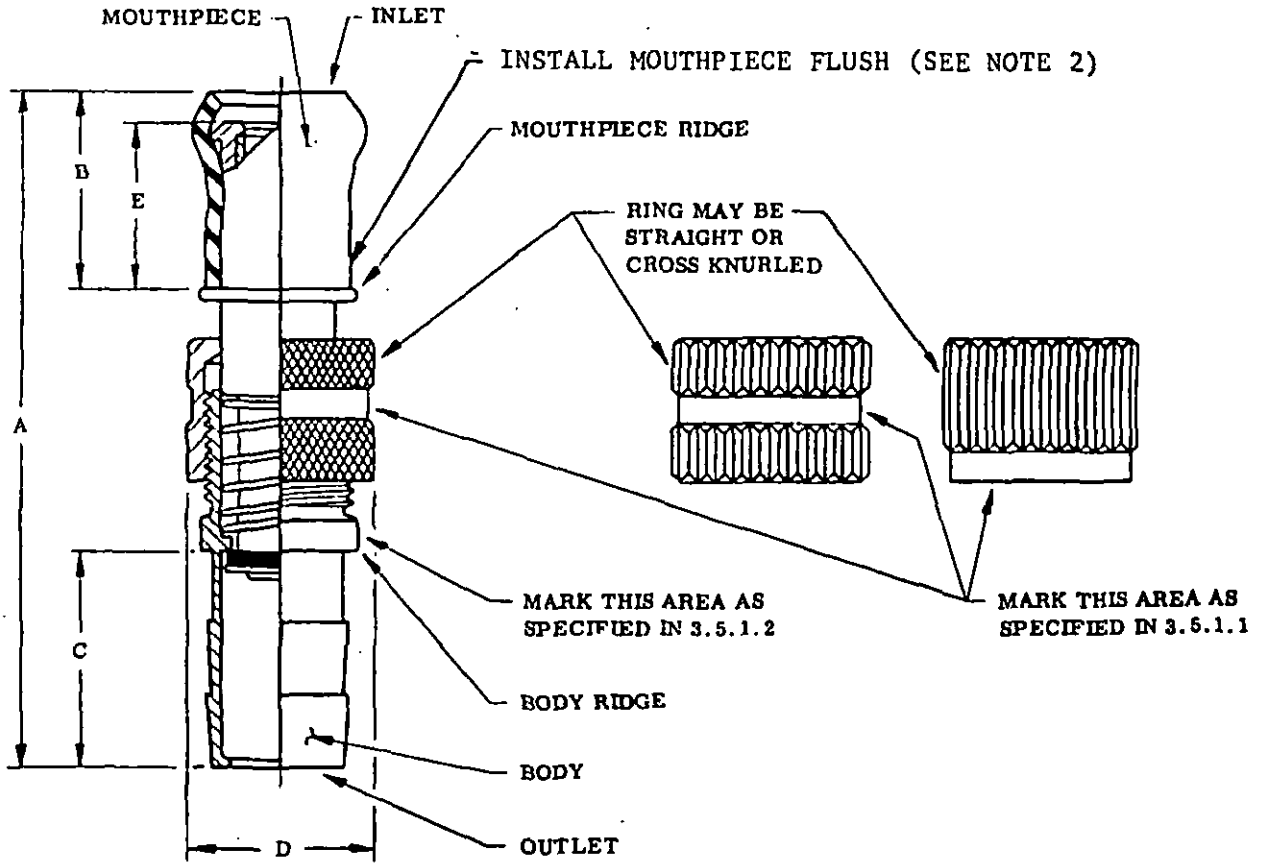
TABLE VII. List of defects for the dimensions of the assembled oral inflation valves, tubes and assemblies (see 4.6.1.1).

Examine	Defect
Measure the oral inflation valves, tubes, and assemblies, as applicable.	Any measurement deviating from the dimensions and tolerances as specified in 3.6, as applicable, shall be enumerated as a dimensional defect.

TABLE VIII. List of defects for packaging (see 4.6.1.2).

Item	Defects
Exterior and Interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers; information on the paper label in the polyethylene bag not readable through the film or no label in the film bag.
Packaging and packing materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the polyethylene bag, unit package, container flaps, loose strappings or inadequately cushioned with the cellulosic material; bulging or distortion of the containers; unit container contains metal fastenings or stitches..
Exterior and Interior weight or content	Gross or net weight exceeds the requirement; more than one class of valves, style of tubes or type assemblies, class of valve and style of tube, as applicable, in the same container; not individually sealed within the polyethylene bag; any portion of the oral inflation valve, tube or assembly, as applicable, distorted, bent or caught in any heat sealed seam.

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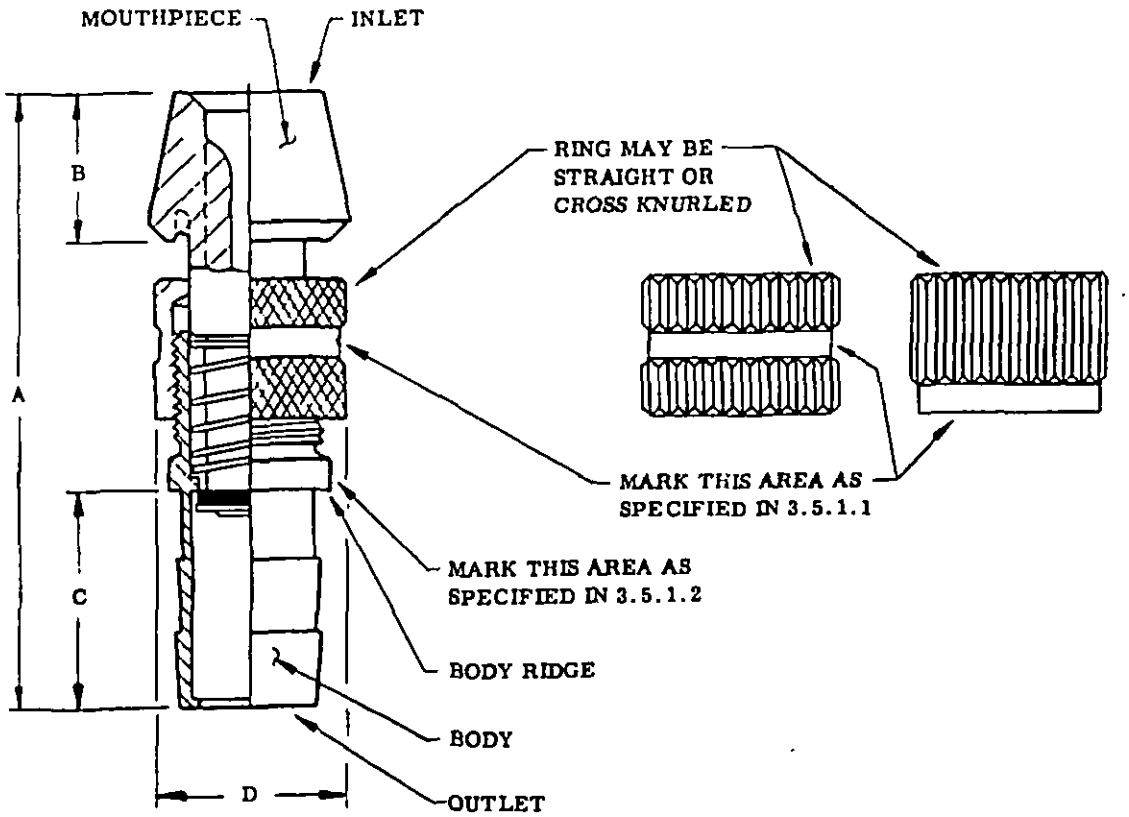
NOTES: 1. THE INTERNAL CONSTRUCTION IS SHOWN FOR INFORMATION ONLY. NO RESTRICTION TO DESIGN IS INTENDED.
2. THE BOTTOM END OF THE RUBBER MOUTHPIECE SHALL BE FLUSH AGAINST THE MOUTHPIECE RIDGE (SEE 3.3.1.2).

ORAL INFLATION VALVE			
	CLASS 1	CLASS 3	
A	2-7/32 to 2-1/2	1-1/2	LENGTH CLOSED
B	3/4 ±1/16	7/16	RUBBER MOUTHPIECE
C	3/4	7/16	SERRATED OUTLET
D	9/16	3/8	BEFORE KNURLING
E	11/16 ±1/16	17/64	METAL OR PLASTIC MOUTHPIECE TUBE

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/64

FIGURE 1. Configuration and dimensions of the class 1 and 3 valves.

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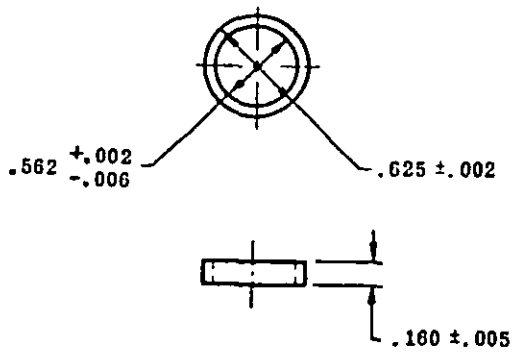
NOTE: THE INTERNAL CONSTRUCTION IS SHOWN FOR INFORMATION ONLY. NO RESTRICTION TO DESIGN IS INTENDED.

ORAL INFLATION VALVE		
	CLASS 2	
A	1-3/4	LENGTH CLOSED
B	1/2	PLASTIC MOUTHPIECE
C	3/4	SERRATED OUTLET
D	5/8	BEFORE KNURLING

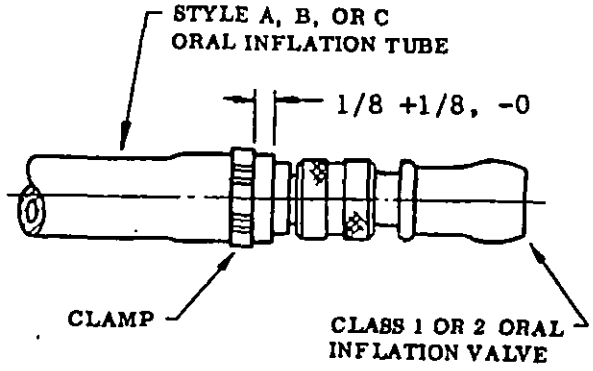
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ±1/64

FIGURE 2. Configuration and dimensions of the class 2 valve.

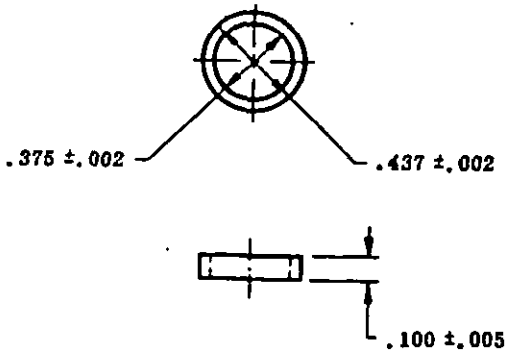
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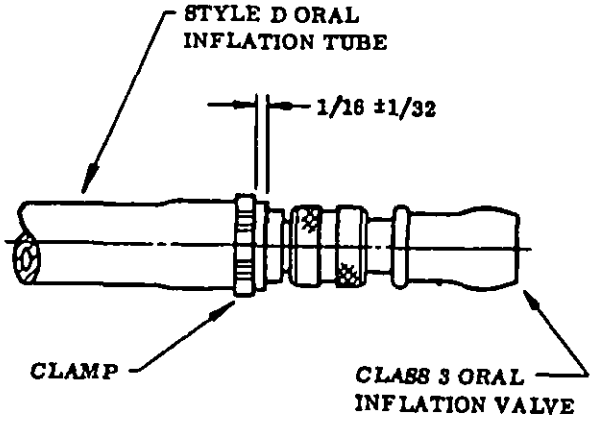
STYLE A, B, AND C ORAL INFLATION TUBE CLAMP



TYPE II ORAL INFLATION ASSEMBLY WITH CLASS 1 OR 2 ORAL INFLATION VALVE AND STYLE A, B, OR C ORAL INFLATION TUBE



STYLE D ORAL INFLATION TUBE CLAMP



TYPE II ORAL INFLATION ASSEMBLY WITH CLASS 3 ORAL INFLATION VALVE AND STYLE D ORAL INFLATION TUBE

DIMENSIONS IN INCHES

FIGURE 3. Type II oral inflation assemblies.

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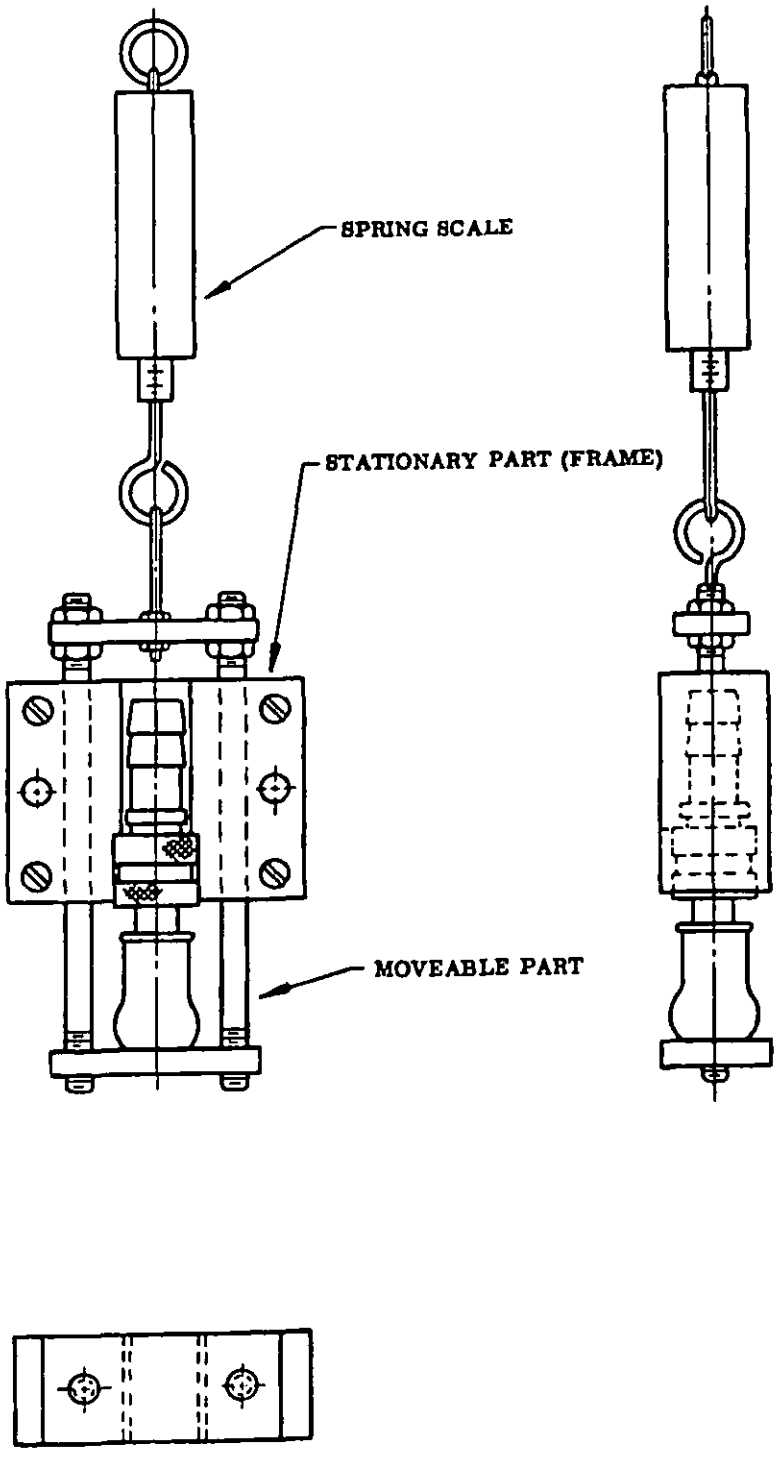
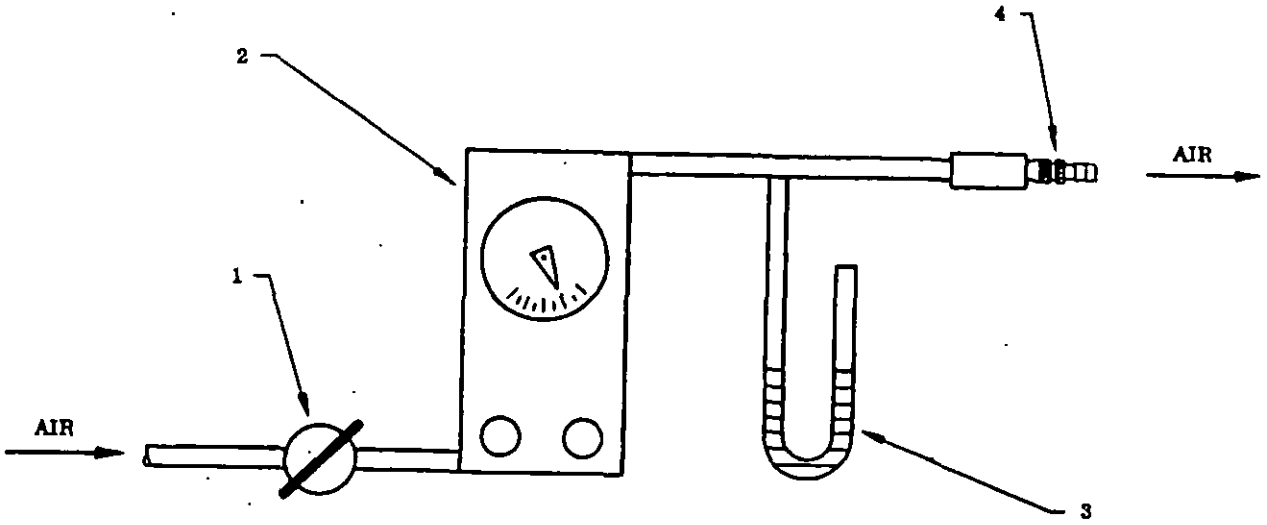


FIGURE 4. Force to open inspection apparatus.

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1. REGULATOR
2. FLOWMETER
3. PRESSURE INDICATOR
4. ORAL INFLATION VALVE (MOUTHPIECE FULLY DEPRESSED)

FIGURE 5. Arrangement of the flow inspection apparatus.

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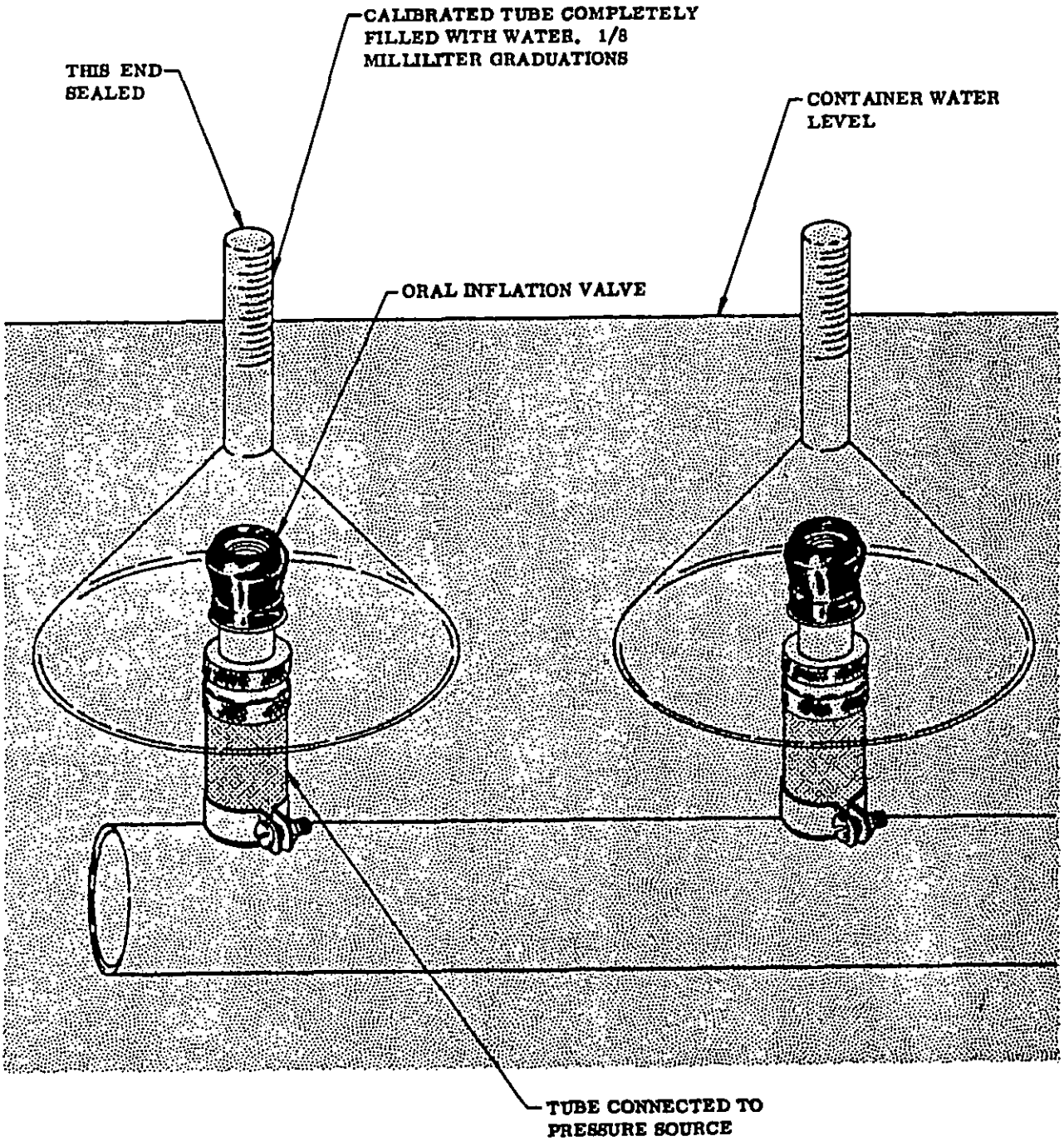


FIGURE 6. Apparatus for determining leakage at 160° F.

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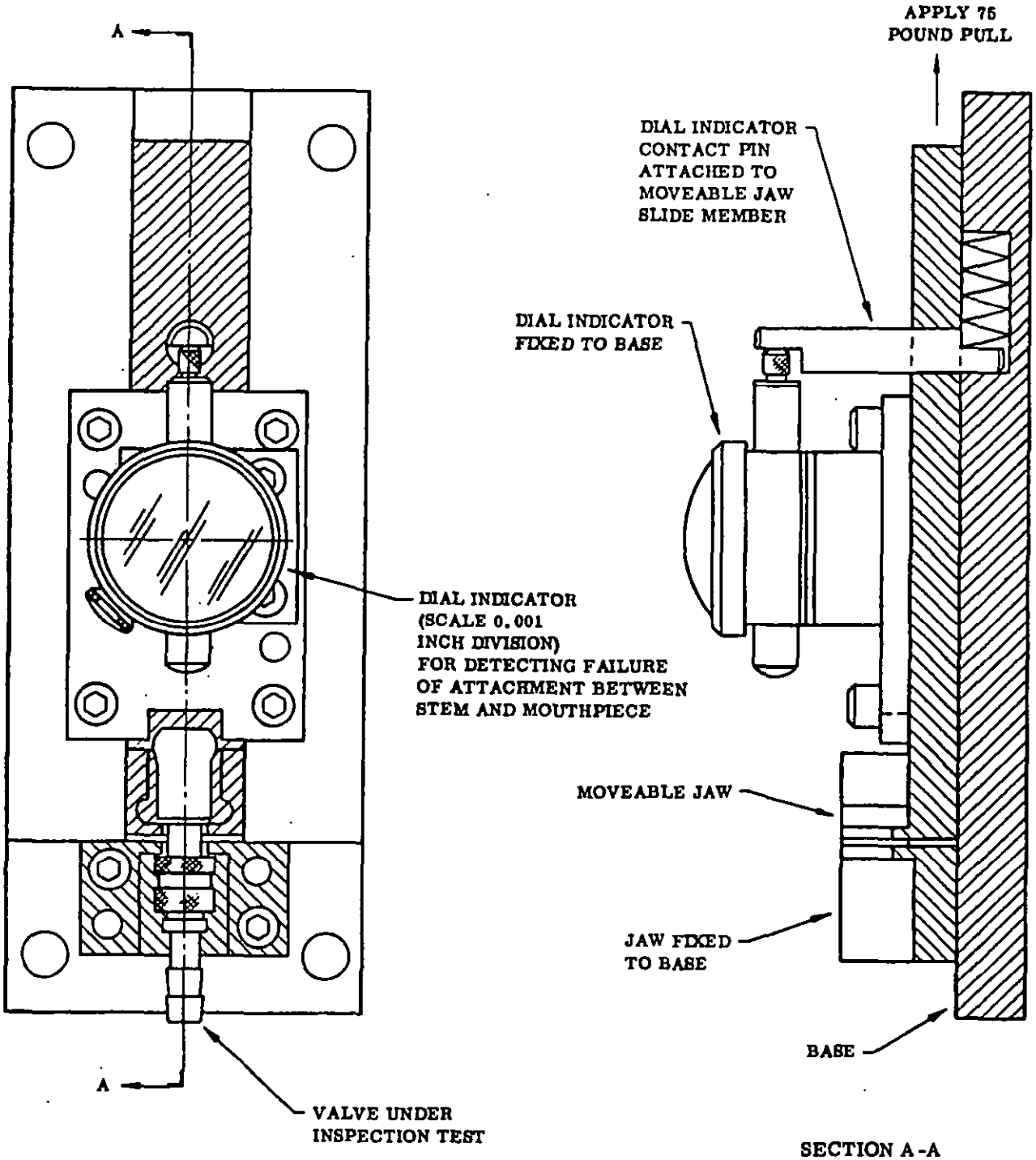


FIGURE 7. Strength of assembly inspection apparatus.

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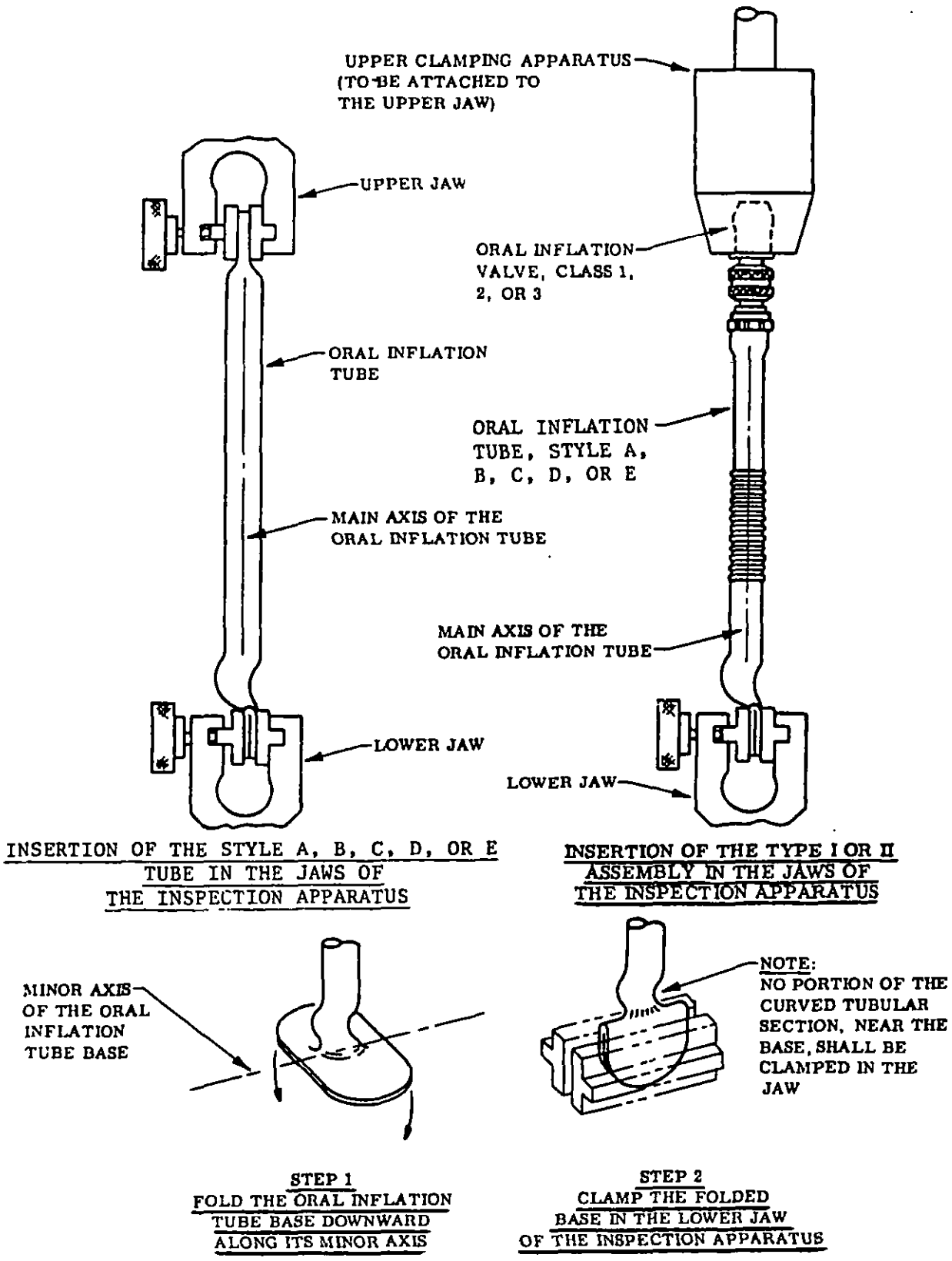


FIGURE 8. Procedure for determining the breaking strength of the oral inflation tube and strength of attachment of the oral inflation tube to the oral inflation valve.

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